

10 (extra): symmetrization

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Nice trick to have See §4.2 Wainwright's "High-Dimensional Statistics" '19
or §6.4 Vershynin's "High-dimensional probability" '18
or §16.2 (Prop. 16.5 "Giné-Zinn" argument) Tropp's ACM 217 lecture notes

Def A r.v. X is **symmetric** if X and $-X$ have the same distribution

Ex: $X \sim N(0, \sigma^2)$ is symmetric

$X = \begin{cases} -1 & \text{wp } 1/2 \\ 1 & \text{wp } 1/2 \end{cases}$ is symmetric $X = \begin{cases} 0 & \text{wp } 1/2 \\ 1 & \text{wp } 1/2 \end{cases}$ not symmetric, $X = \begin{cases} -1 & \text{wp } 1/3 \\ 1 & \text{wp } 2/3 \end{cases}$ not symmetric
recall we call this "Rademacher"

Ex: let $X \sim D$, and $X' \sim D$, then $Y = X - X'$ (over $D \times D'$)
is always symmetric!

Ex: Let $Z \sim \text{Rademacher}$, $X \sim D$. Then $Z \cdot X$ and $Z \cdot |X|$ both symmetric and in fact have the same distribution (and same as X if X is also symmetric)
and independent
i.e. $Z \cdot (X - X') \sim (X - X')$

Lemma 6.1.2 in Vershynin If Y and Z are independent and $\mathbb{E}[Z] = 0$,
then for any convex function f , $\mathbb{E}f(Y) \leq \mathbb{E}f(Y+Z)$

Proof

$$\begin{aligned} \mathbb{E}f(Y+Z) &= \mathbb{E} \mathbb{E}(f(Y+Z) | Y) && \text{tower property} \\ &\geq \mathbb{E} f(\mathbb{E}(Y+Z | Y)) && \text{Jensen's on } \mathbb{E}(\cdot | Y) \\ &= \mathbb{E} f(Y + \mathbb{E}[Z | Y]) && \text{linearity of } \mathbb{E} \\ &= \mathbb{E} f(Y + \mathbb{E}[Z]) && \text{independence} \\ &= \mathbb{E} f(Y) && \text{mean 0} \quad \square \end{aligned}$$

Application: bounded r.v. are sub-Gaussian cf. Math 888 lecture notes of Sébastien Roch

Prop: Let $\mathbb{E}X = 0$, $|X| \leq c$ a.s., then $\mathbb{E}e^{\lambda X} \leq e^{Zc^2\lambda^2} \quad \forall \lambda \in \mathbb{R}$

proof Let $Z \sim \text{Rademacher}$ and X' an independent copy of X . Since $x \mapsto e^{\lambda x}$ is convex,
via Lemma, $\mathbb{E}_X e^{\lambda X} \leq \mathbb{E}_{X, X'} e^{\lambda(X - X')} = \mathbb{E}_{X, X', Z} e^{\lambda Z(X - X')}$ *Since same distribution*

FACT: MGF of Rademacher

$$\mathbb{E} e^{\mu Z} = \cosh(\mu) \leq e^{\mu^2/2}$$

$$\text{since } e^a = \sum_{n=0}^{\infty} \frac{a^n}{n!}$$

$$\frac{1}{2}(e^a + e^{-a}) = \frac{1}{2} \sum_{n=0}^{\infty} \frac{1}{n!} (a^n + (-a)^n) = \sum_{k=0}^{\infty} \frac{a^{2k}}{(2k)!} \quad \text{only even terms remain}$$

$$e^{a^2/2} = \sum_{m=0}^{\infty} \frac{a^{2m}}{2^m m!} \quad \text{and } (2n)! \geq 2^n n!$$

$$\begin{aligned} &= \mathbb{E}_{X, X'} \mathbb{E}[e^{\lambda Z(X - X')} | X, X'] && \text{tower property} \\ &\leq \mathbb{E}_{X, X'} e^{\frac{1}{2} \lambda^2 (X - X')^2} && \text{via MGF of Rademacher} \\ &\leq e^{Zc^2\lambda^2} && \text{since } |X|, |X'| \leq c \end{aligned}$$

See also Lemma D.1
Hoeffding's lemma
 $\mathbb{E}[X] = 0, a \leq X \leq b$ a.s.
then $\forall \lambda$
 $\mathbb{E}[e^{\lambda X}] \leq e^{(b-a)^2 \lambda^2 / 8}$
so if $a = -b = c$
get $\frac{1}{2} c^2$,
factor of 4 better

(p. 2: symmetrization)

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Another application *Prop 4.11 Wainwright*
Thm D.9 Mohri et al. is similar, Khintchine-Kahane ineq.

Lemma 6.4.2 "Symmetrization" in Vershynin $X_1, \dots, X_n$ independent, mean 0 random vectors,

$$\text{then } \frac{1}{2} \mathbb{E} \left\| \sum_{i=1}^n \varepsilon_i X_i \right\| \stackrel{\textcircled{B}}{\leq} \mathbb{E} \left\| \sum_{i=1}^n X_i \right\| \stackrel{\textcircled{A}}{\leq} 2 \mathbb{E} \left\| \sum_{i=1}^n \varepsilon_i X_i \right\|$$

where ε_i are independent Rademachers.

Proof Extending our lemma 6.1.2 to vectors and using that $\vec{x} \mapsto \|\vec{x}\|$ is convex, we get

$$\begin{aligned} \textcircled{A} \text{ } P &:= \mathbb{E} \left\| \sum X_i \right\| \stackrel{\text{mean 0 by assumption}}{\leq} \mathbb{E} \left\| \sum X_i - \sum X_i' \right\| \\ &= \mathbb{E} \left\| \sum (X_i - X_i') \right\| \quad \text{symmetrization trick} \\ &= \mathbb{E} \left\| \sum \varepsilon_i (X_i - X_i') \right\| \\ &\leq \mathbb{E} \left\| \sum \varepsilon_i X_i \right\| + \mathbb{E} \left\| \sum \varepsilon_i X_i' \right\| \quad \text{triangle ineq.} \\ &= 2 \mathbb{E} \left\| \sum \varepsilon_i X_i \right\| \end{aligned}$$

$\textcircled{B}$ similar:

$$\begin{aligned} \mathbb{E} \left\| \sum \varepsilon_i X_i \right\| &\leq \mathbb{E} \left\| \sum \varepsilon_i (X_i - X_i') \right\| \quad \text{lemma 6.1.2 again, } \mathbb{E}[\varepsilon_i X_i'] = 0 \text{ still} \\ &= \mathbb{E} \left\| \sum (X_i - X_i') \right\| \quad \text{symmetrization argument, in reverse} \\ &\leq \mathbb{E} \left\| \sum X_i \right\| + \mathbb{E} \left\| \sum X_i' \right\| = 2 \mathbb{E} \left\| \sum X_i \right\| \quad \square \end{aligned}$$

Also used for Hanson-Wright ineq.